

Intro to Learning-Augmented Algorithms (Algorithm with Predictions)

Adapted from Alex Andoni, “Lecture 24: Learning Augmented Algorithms”, scribed by Kiran Vodrahalli and Lei You,
<https://www.mit.edu/~andoni/algoS19/scribes/scribe24.pdf>

Algorithms with a fallible predictor

Ordinary input + predicted advice → algorithm

THE PREDICTOR

A model or heuristic predicts something useful about this instance.

THE ALGORITHM

Uses the forecast to guide decisions. We analyze what happens when it is wrong.

The performance guarantee refers to prediction quality, not how the predictor was trained.

Competitive Ratio

For one instance I :

$$r_a(I) = \frac{Cost_A(I)}{OPT(I)}$$

$OPT(I)$ knows the complete input, including the future.

Worst case: take the largest ratio over allowed instances.

For costs, a ratio of 1 is optimal.

Three ways to judge the advice

CONSISTENCY

Good performance with an exact prediction ($e = 0$)

ROBUSTNESS

A guarantee that holds for every prediction

SMOOTHNESS

A bound that changes gradually with the error e

What is an online algorithm?

REVEALED OVER TIME

The algorithm sees past and present inputs, but not future ones.

DECIDE BEFORE THE FUTURE

Its current action may be impossible to undo.

OFFLINE BENCHMARK

OPT makes its decisions knowing the whole input sequence.

Online problems are a natural starting point

A prediction estimates information unavailable when the decision is due.

BENEFIT

Better decisions if the forecast is accurate.

RISK

A poor forecast may cause an irreversible costly action.

Buy or rent: the online problem

Rent: \$1 per day

Buy: \$b once

Actual number of use days x is unknown in advance.

Offline optimum: $OPT(x) = \min\{x, b\}$

The Deterministic Baseline

Rent for $b - 1$ days; buy on day b if still needed.

$$x < b$$

$$\text{Cost} = x; \text{ratio} = 1$$

$$x \geq b$$

$$\text{Cost} = (b - 1) + b = 2b - 1$$

$$\text{Competitive ratio} = 2 - \frac{1}{b} < 2$$

Claim: This is optimal for a deterministic algorithm.

Online algorithm with predictor

Predictor forecasts the number of use days: y Error $e = |x - y|$

DEFINITION: α -CONSISTENCY

If $y = x$ ($e = 0$), competitive ratio $\leq \alpha$ for every actual x .

DEFINITION: β -ROBUSTNESS

For every actual x and every prediction y , competitive ratio $\leq \beta$.

Smaller α and β are better; achieving both may require a tradeoff.

Algorithm 1: trust the predictor completely

PREDICTED LONG SEASON: $y \geq b$

Buy immediately for cost b .

PREDICTED SHORT SEASON: $y < b$

Keep renting every day actually used.

With a perfect prediction, this makes the offline-optimal choice.

Performance analysis

Cost of Algorithm 1 $\leq OPT(x) + e$

ACTUAL x AND FORECAST y ON SAME SIDE OF b

The buy/rent choice matches OPT; extra cost = 0.

$y \geq b > x$: BUY TOO EARLY

Extra cost $b - x \leq y - x = e$.

$x \geq b > y$: RENT TOO LONG

Extra cost $x - b \leq x - y = e$.

Consistency and robustness

PERFECT PREDICTOR: $e = 0$

Ratio = 1, so Algorithm 1 is 1-consistent.

BAD PREDICTION: $e \rightarrow \infty$

Rent cost x ; $\text{OPT} = b$. Ratio x/b has no finite cap.

Ratio $\leq 1 + e / \text{OPT}(x)$

An error-dependent bound alone need not give robustness.

Algorithm 2: learning-augmented version

Choose λ in $(0, 1]$ to control trust before seeing the true x .

PREDICTOR FORECASTS LONG: $y \geq b$

Rent and buy on day $\lceil \lambda b \rceil$ if still needed.

PREDICTOR FORECASTS SHORT: $y < b$

Rent and buy on day $\lceil b/\lambda \rceil$ if still needed.

Smaller λ makes each decision follow the forecast more strongly.

$\lambda = 1$ reduces Algorithm 2 to the deterministic baseline

$\lambda \rightarrow 0$ reduces Algorithm 2 to Algorithm 1

Performance analysis

$$\text{Competitive ratio} \leq \min \left\{ \frac{1+\lambda}{\lambda}, 1 + \lambda + \frac{e}{(1-\lambda)OPT(x)} \right\}$$

Prediction and reality	What happens
$y \geq b, x \geq b$	Buying early costs at most $\lambda b + b = (1 + \lambda)OPT(x)$.
$y < b, x < b$	The algorithm never buys, so its cost equals $OPT(x) = x$.
$y \geq b > x$	If actual period ends before the buying day, cost is x . Otherwise, cost $\leq (1 + \lambda)b$. Since $e = y - x \geq b - x$, and $1 + \lambda < \frac{1}{1-\lambda}$, we obtain cost $\leq (1 + \lambda)OPT(x) + \frac{e}{1-\lambda}$.
$x \geq b > y$	If actual period ends before the buying day, cost = $x \leq \frac{b}{\lambda} = \frac{OPT(x)}{\lambda}$. Otherwise, cost $\leq \frac{b}{\lambda} + b = \left(\frac{1+\lambda}{\lambda}\right)OPT(x)$.

Performance analysis

$$\text{Competitive ratio} \leq \min \left\{ \frac{1+\lambda}{\lambda}, 1 + \lambda + \frac{e}{(1-\lambda)OPT(x)} \right\}$$

CONSISTENCY

$e = 0$: ratio $\leq 1 + \lambda$

ROBUSTNESS

Arbitrarily large e : ratio $\leq 1 + \frac{1}{\lambda}$

SMOOTHNESS

The second term varies with e .

Choosing λ : how much to trust the predictor

λ	PERFECT-PREDICTION BOUND	ANY-PREDICTION BOUND
0.2	1.2	6
0.5	1.5	3
0.8	1.8	2.25

Small λ : more trust, better consistency, weaker worst-case guarantee.

Large λ : less trust, stronger worst-case guarantee.